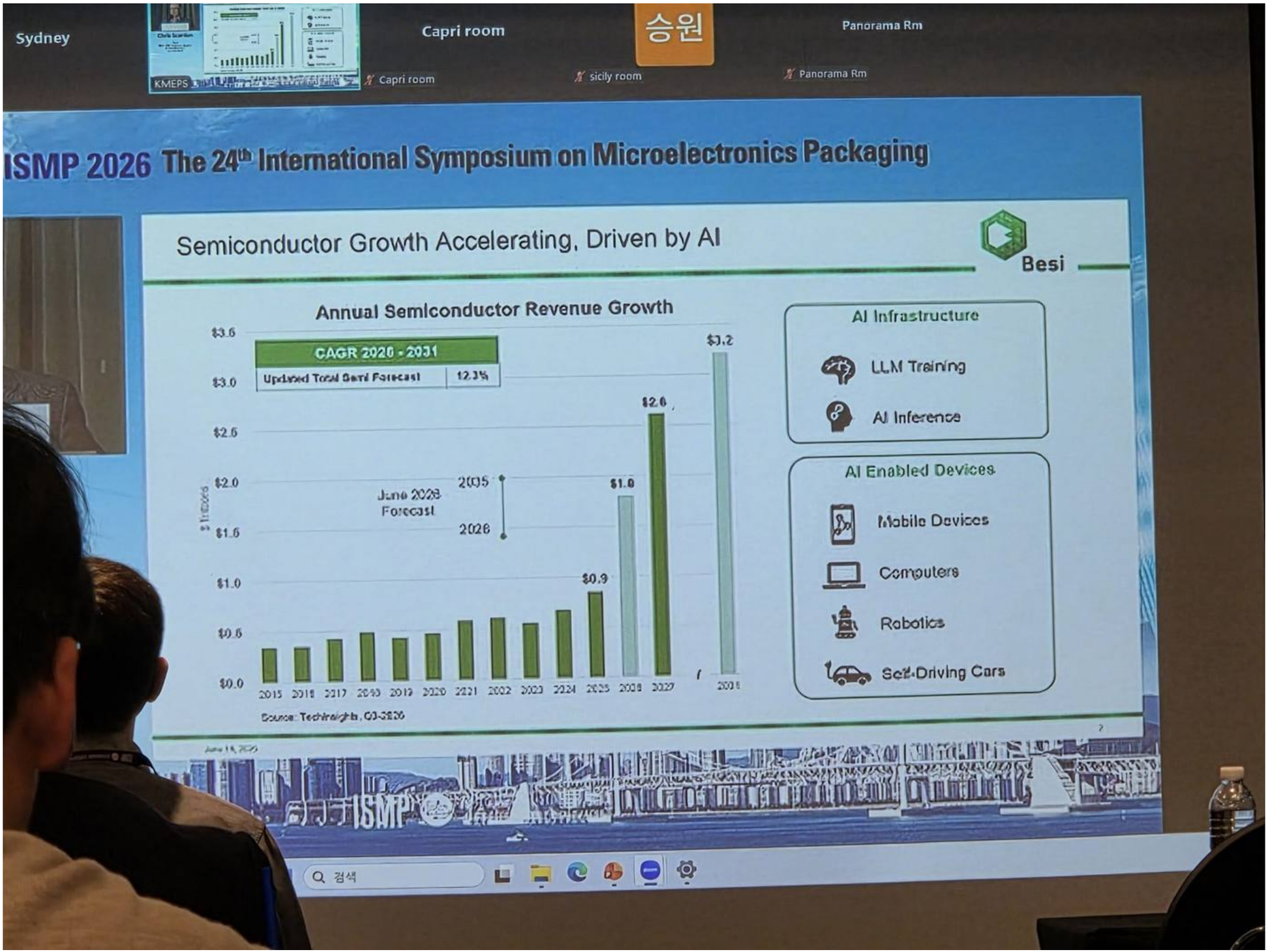
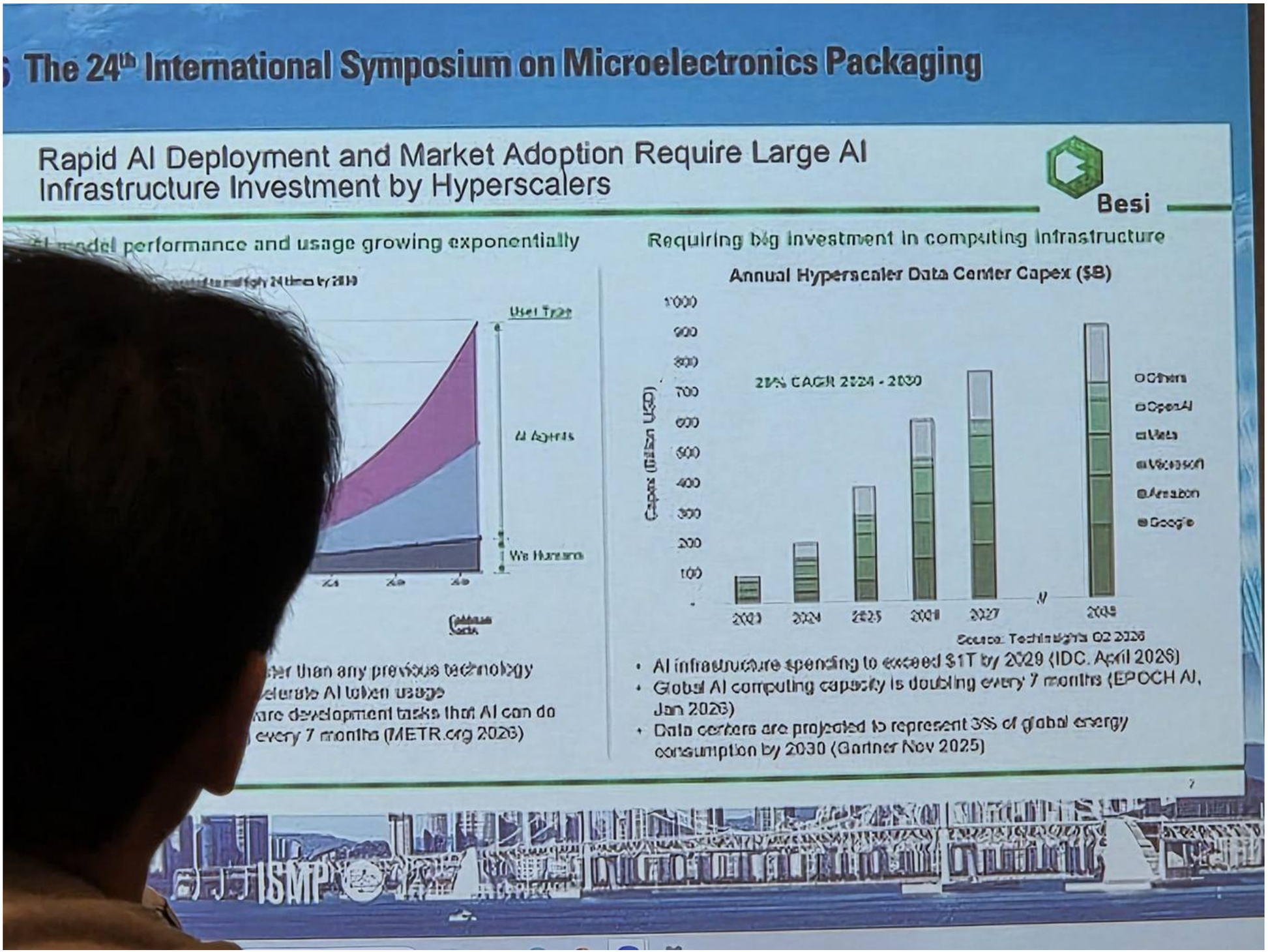
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



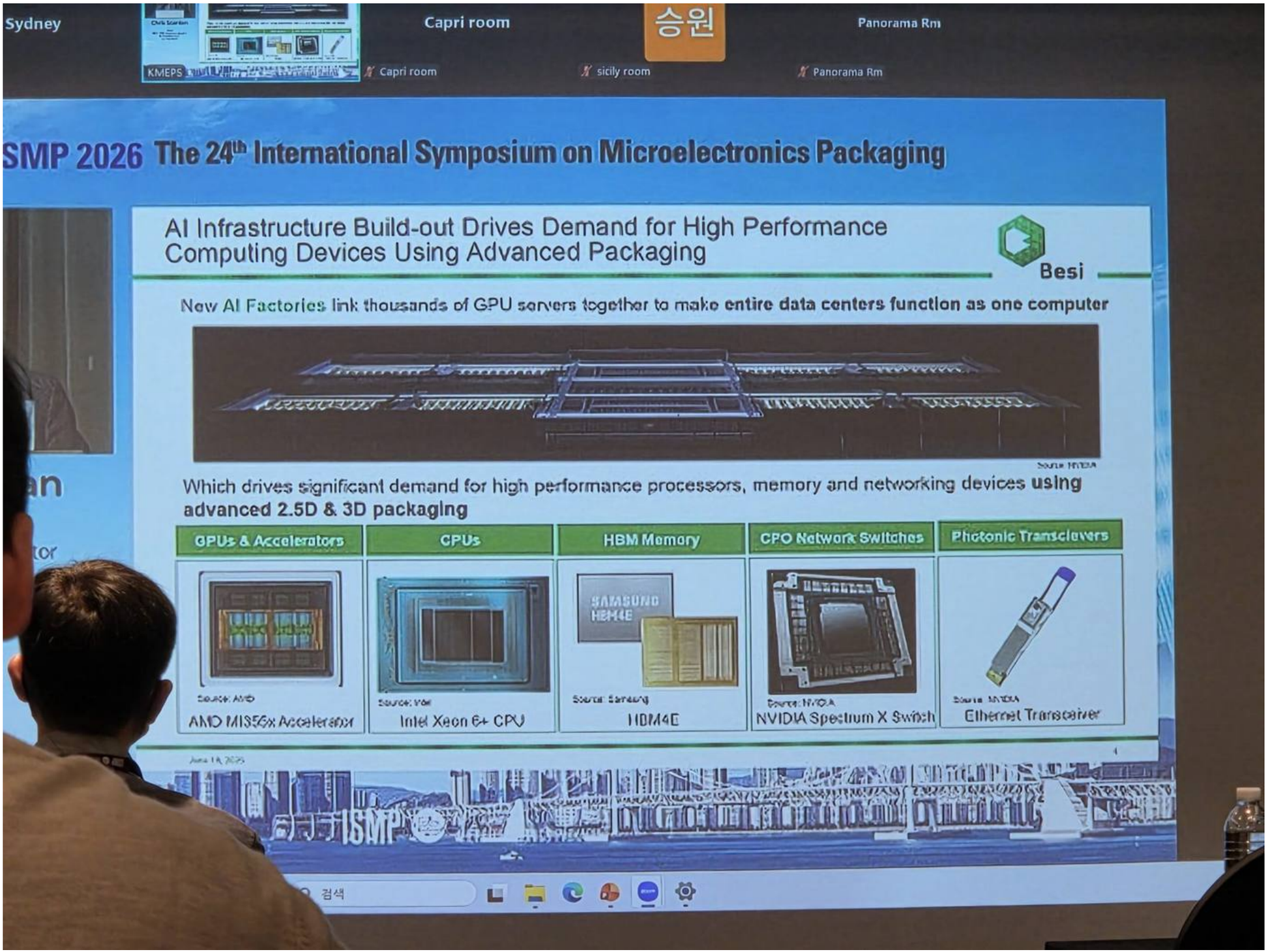
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



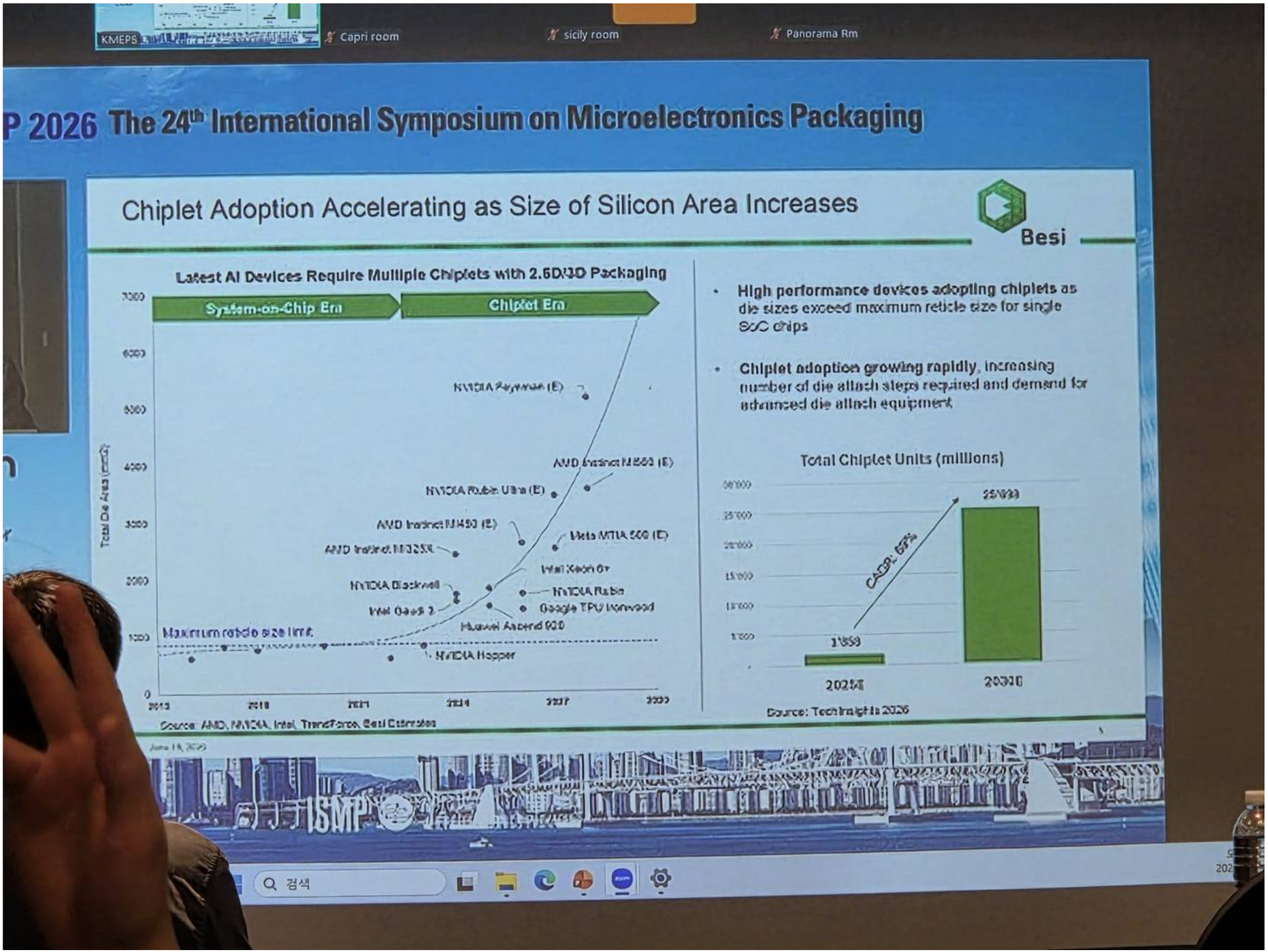
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



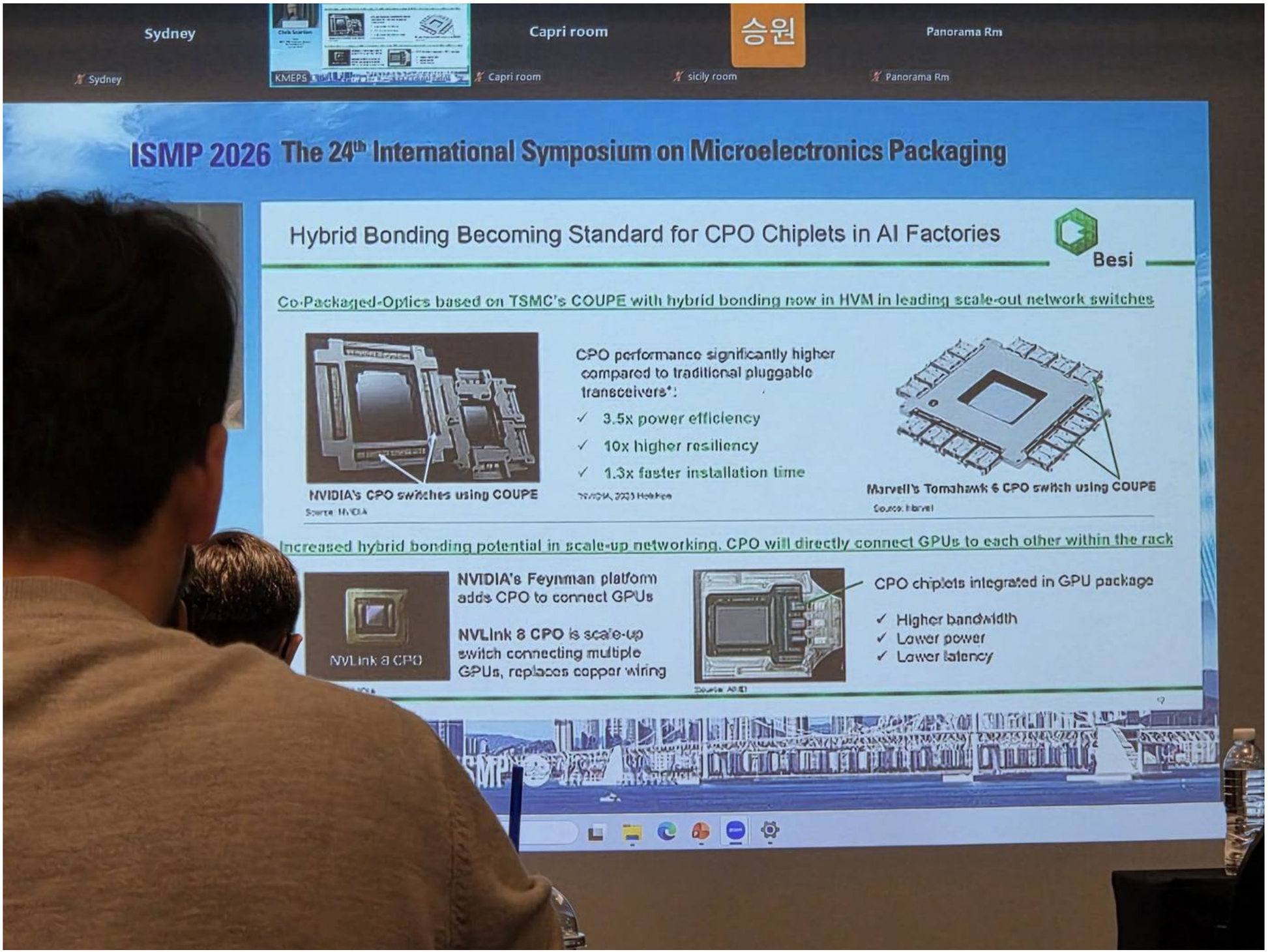
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



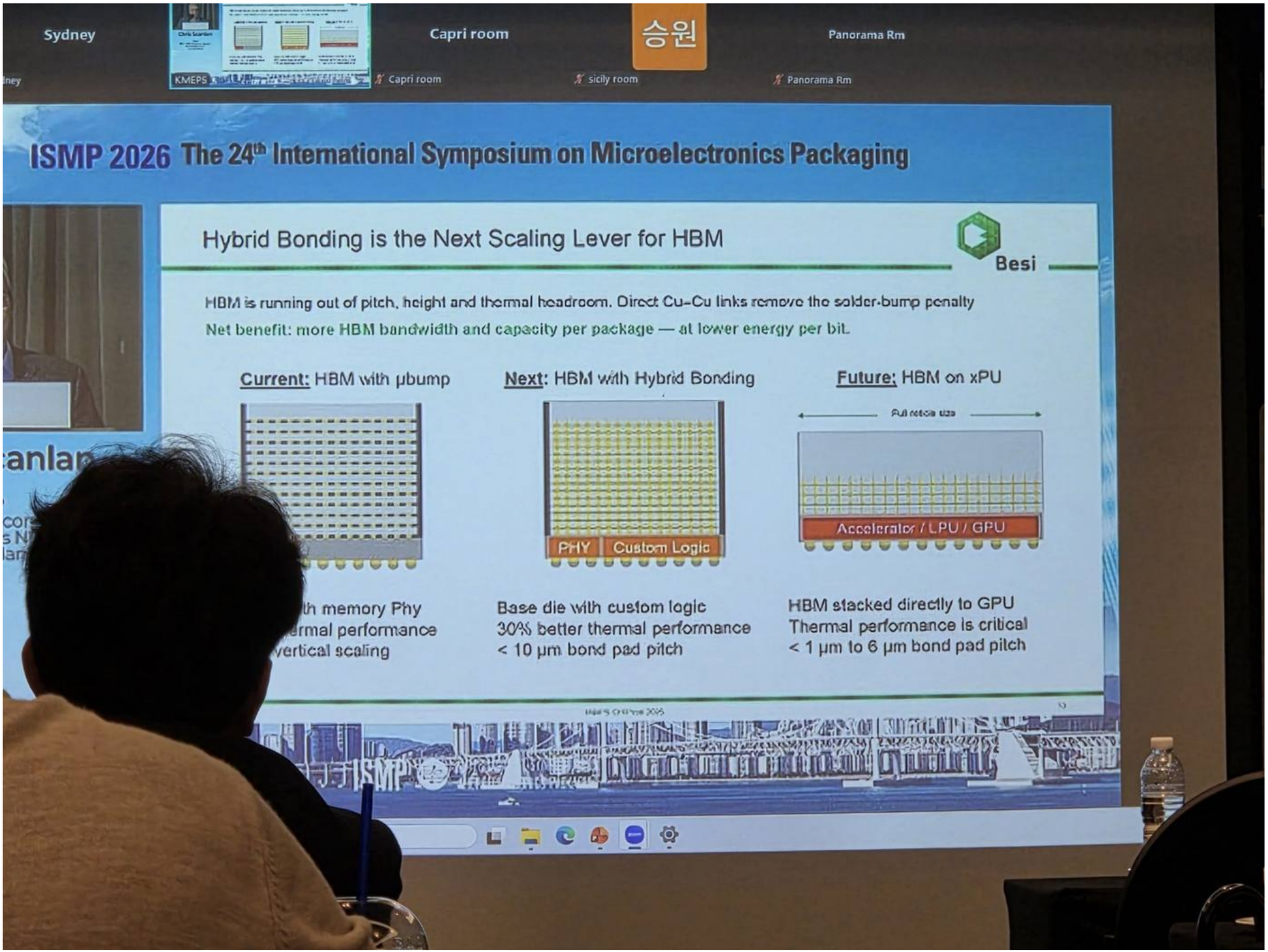
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



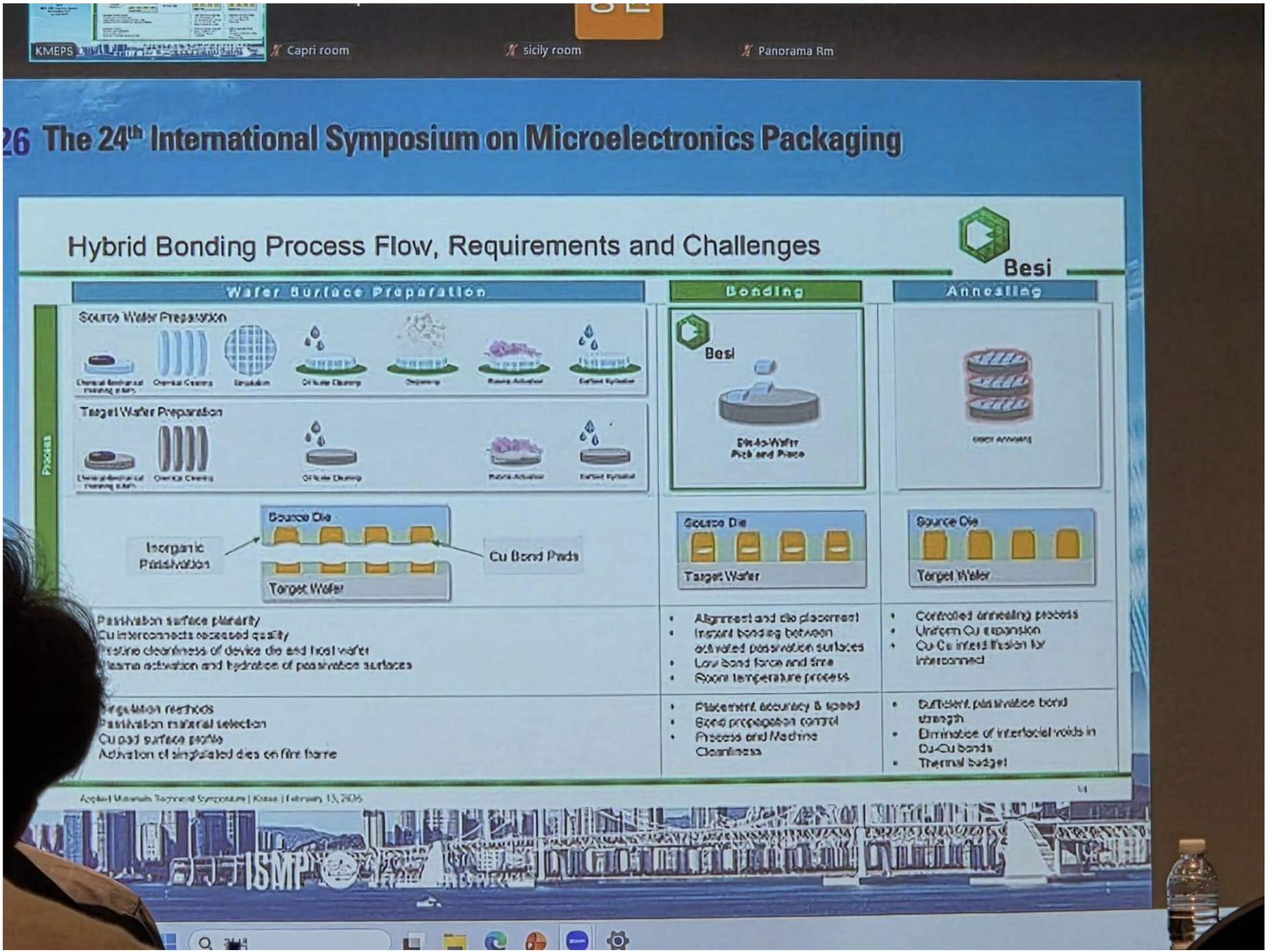
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



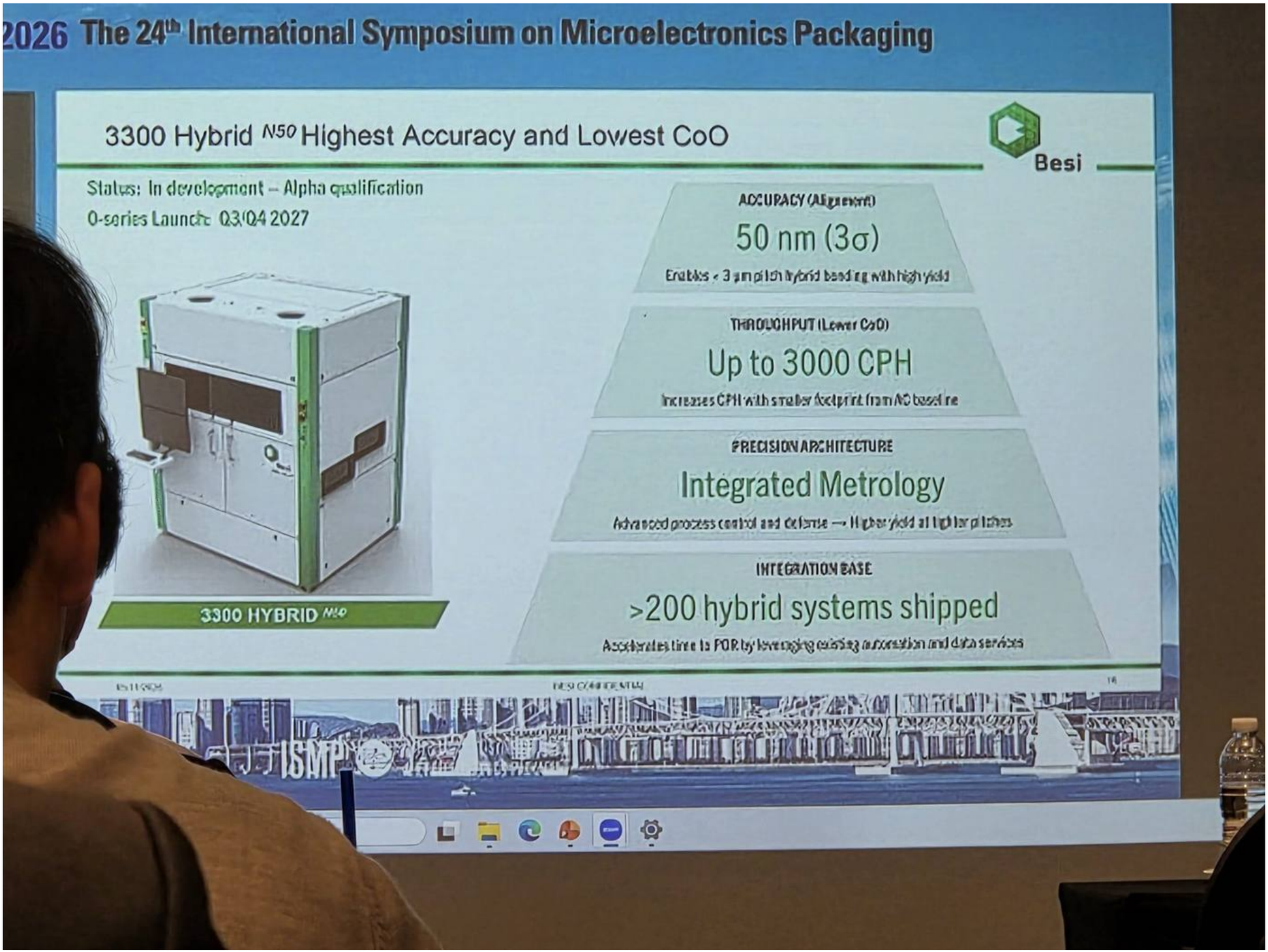
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



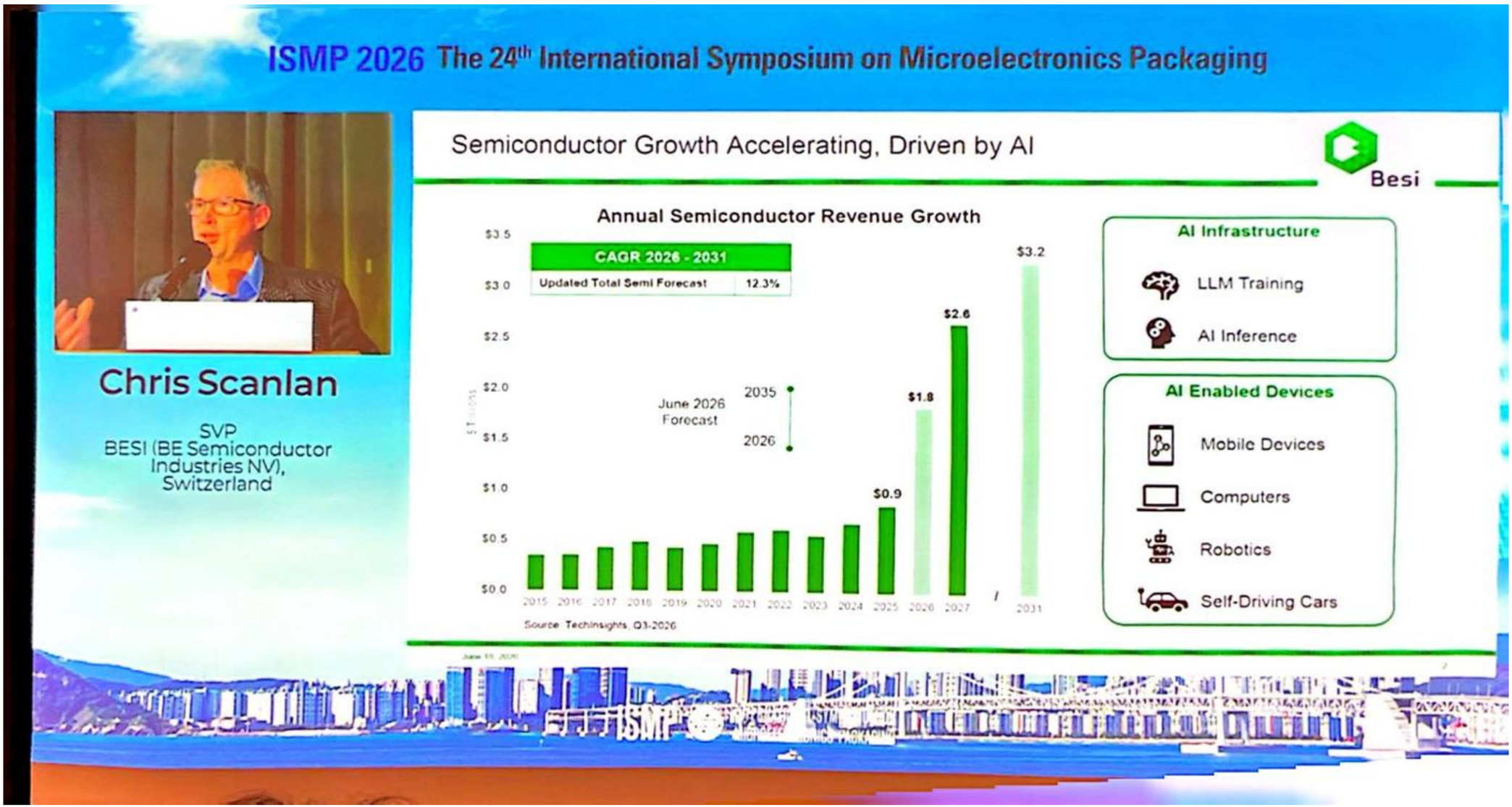
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi

ISMP 2026 The 24th International Symposium on Microelectronics Packaging



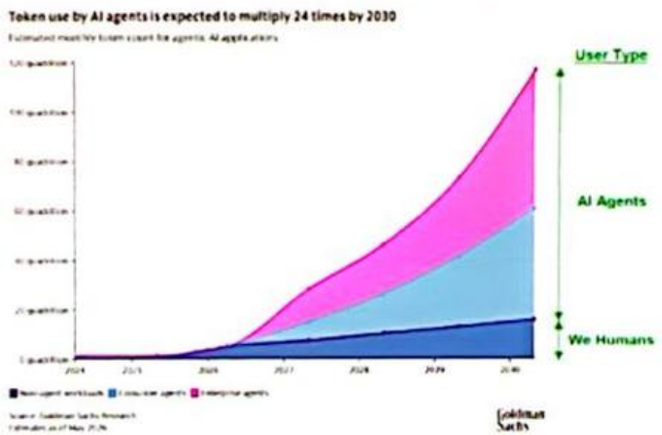
Chris Scanlan

SVP
BESI (BE Semiconductor Industries NV), Switzerland

Rapid AI Deployment and Market Adoption Require Large AI Infrastructure Investment by Hyperscalers

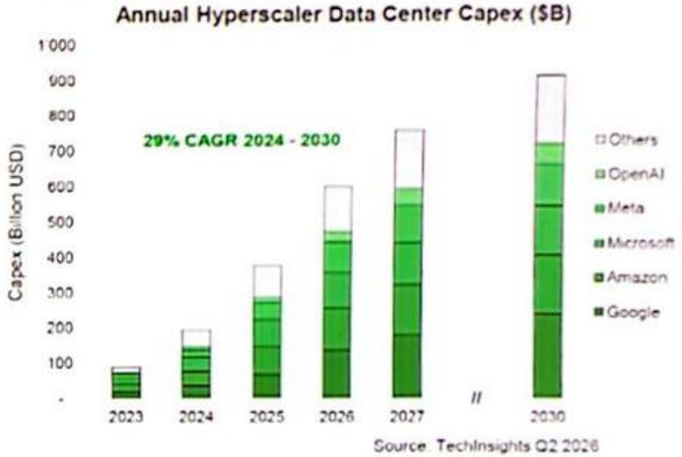
Besi

AI model performance and usage growing exponentially



- Generative AI adopted faster than any previous technology
- Automated AI agents accelerate AI token usage
- Human duration of software development tasks that AI can do successfully is doubling every 7 months (METR org 2026)

Requiring big investment in computing infrastructure

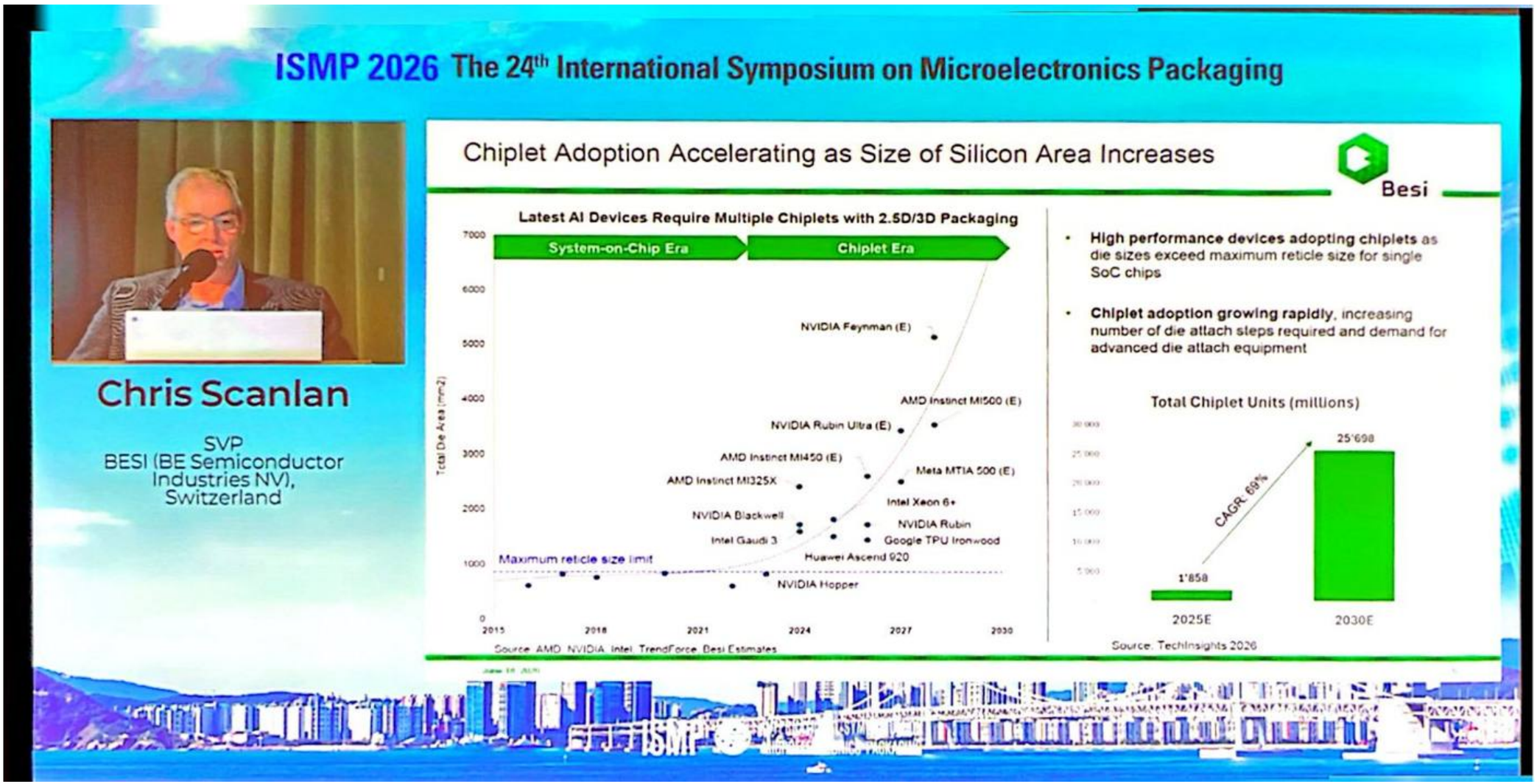


Source: TechInsights Q2 2026

- AI infrastructure spending to exceed $1T by 2029 (IDC, April 2026)
- Global AI computing capacity is doubling every 7 months (EPOCH AI, Jan 2026)
- Data centers are projected to represent 3% of global energy consumption by 2030 (Gartner Nov 2025)

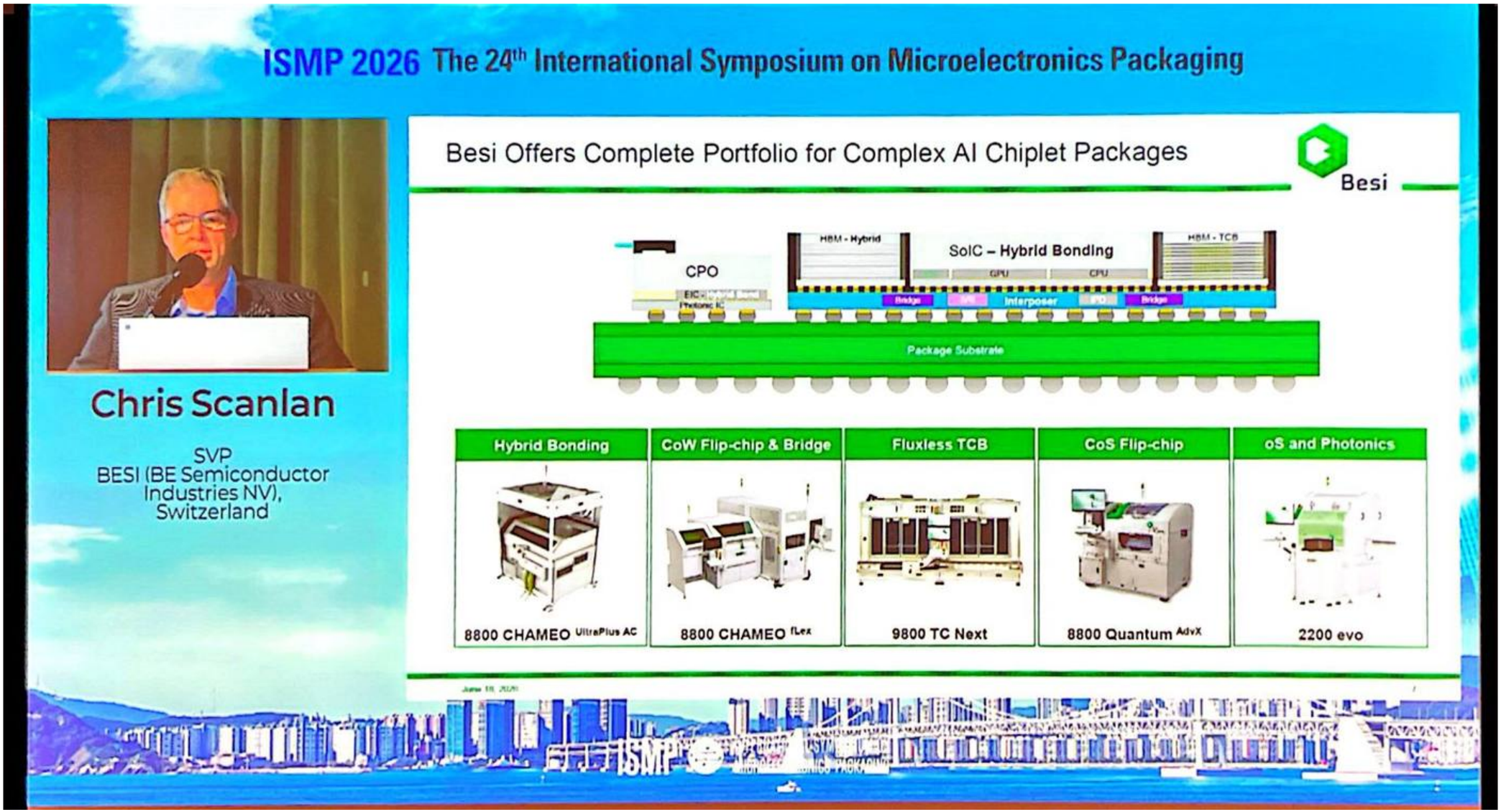
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi

ISMP 2026 The 24th International Symposium on Microelectronics Packaging



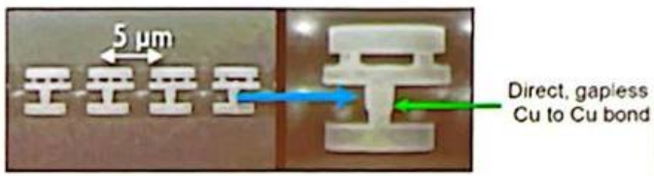
Chris Scanlan

SVP
BESI (BE Semiconductor Industries NV), Switzerland

Hybrid Bonding is Interconnect Technology of Choice for High Performance 3D Heterogeneous Integration

Besi

Hybrid Bonding is extension of front-end fab process
Directly connects chips in 3D with highest interconnect density



Source: AMD

TCB uses µ-bump interconnect to connect chips with solder and epoxy encapsulation materials

Solder µbumps with gap fill material



- Lower bump density
- Higher thermal resistance
- Back-end assembly process

Source: Intel

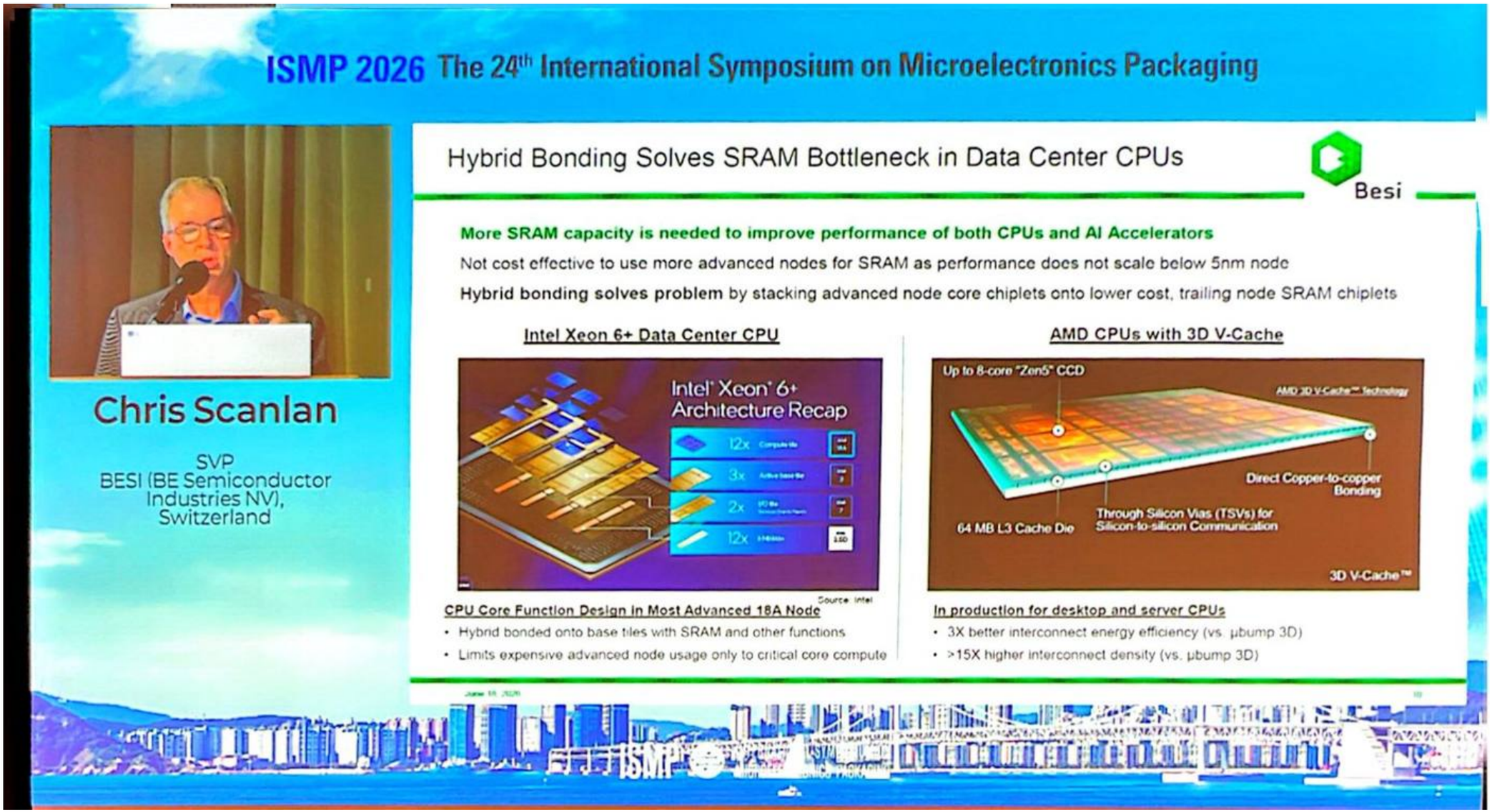
Hybrid Bonding Benefits for 3D Chip Stacking

- ✓ 15X Higher Interconnect Density[1]
- ✓ 11.9X Higher Speed[1]
- ✓ 191X Higher Bandwidth Density[1]
- ✓ >100X Higher Energy Efficient Performance[2]
- ✓ Up to 30% Lower Thermal Resistance[3]
- ✓ Higher Reliability Performance[4]
- ✓ 10X Lower Cost per Interconnect[4]

Notes: 1) TSMC, 2023 ECTC, 2) TSMC 2025 ECTC, 3) Samsung 2026 ECTC, 4) Besi Company Analysis

T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi

ISMP 2026 The 24th International Symposium on Microelectronics Packaging



Chris Scanlan

SVP
BESI (BE Semiconductor Industries NV), Switzerland

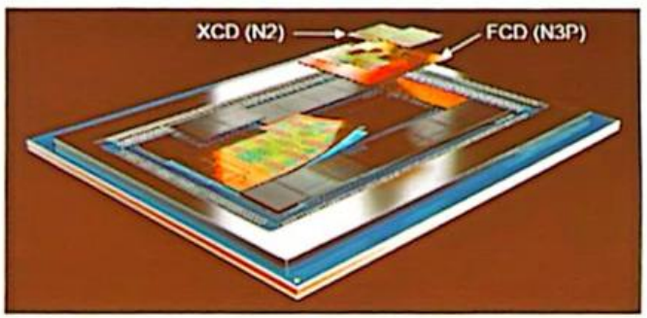
Increased Use of Hybrid Bonding in GPUs and Accelerators Expected

Besi

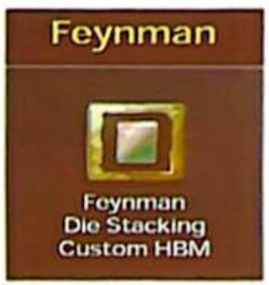
AMD expanding use of hybrid bonding in AI GPUs

- MI355X with 6 µm pitch hybrid bonding now in HVM
- MI455X with larger die sizes introduced
- Stacking of XCD accelerator chiplets to FCD (I/O Fabric and Cache memory) die using hybrid bonding
- Higher compute density and performance per watt

AMD MI455X GPU



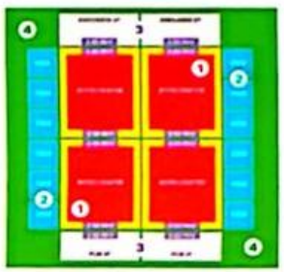
Source AMD



Source NVIDIA

NVIDIA's next generation Feynman (2028) has multiple potential hybrid bonding use cases

- Feynman GPU with 3D die stacking
- Introduction of customized HBM



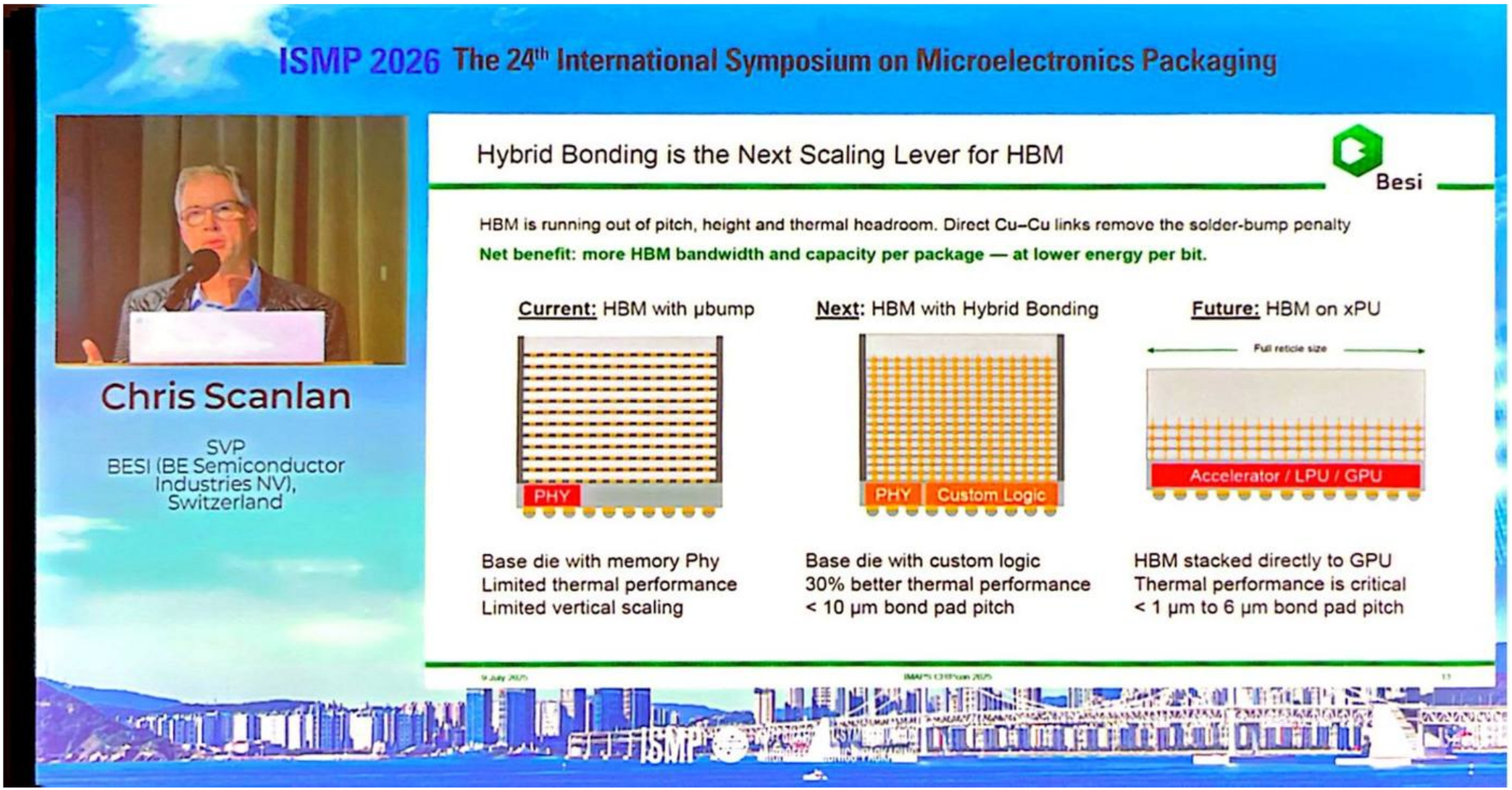
Source Broadcom

Broadcom's 3.5D XDSiP

- Hybrid bonding of accelerator chiplets to I/O and SRAM die
- Accelerators also benefit from additional SRAM capacity and bandwidth

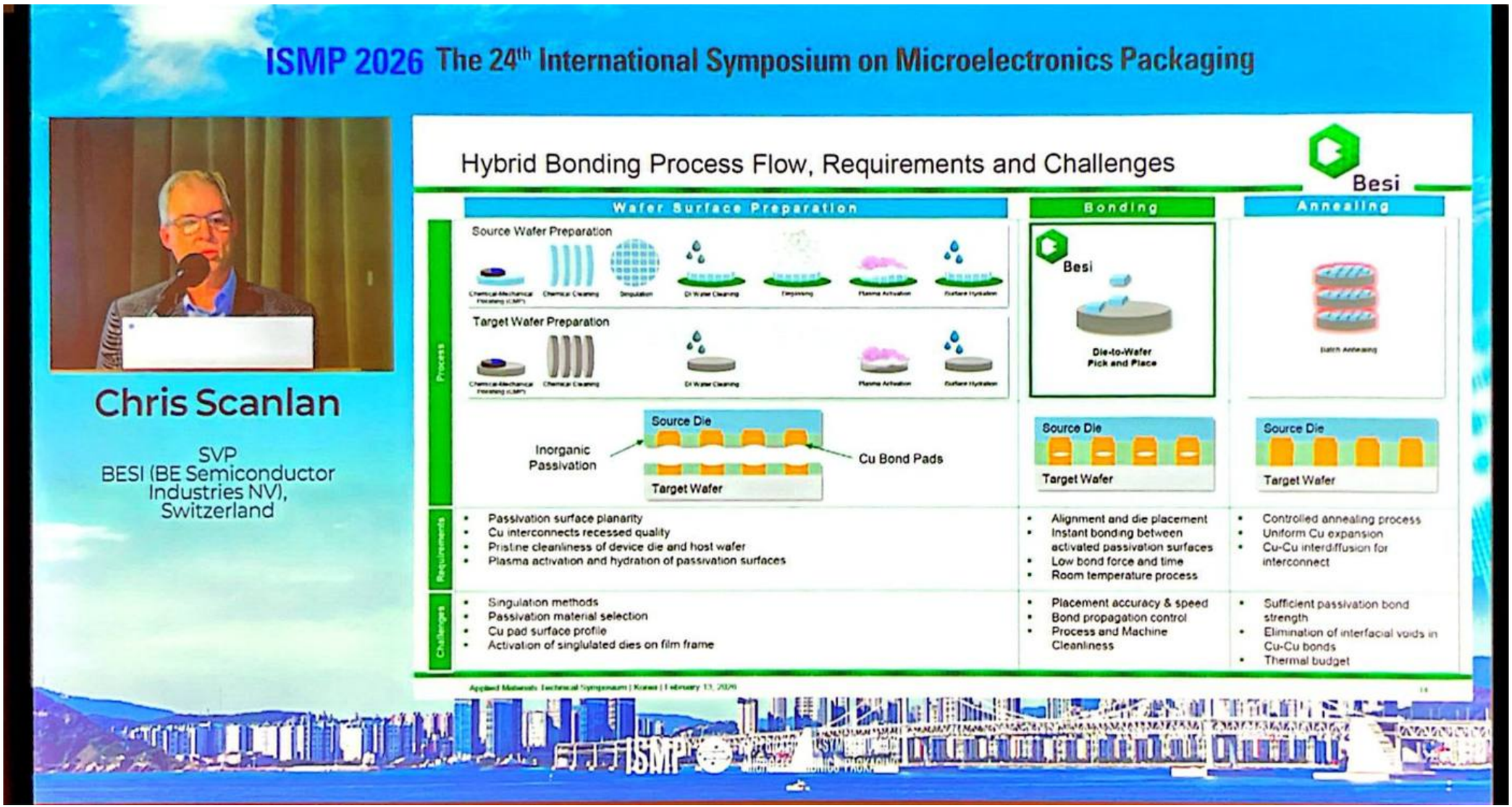
T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



T20 하이브리드 본딩의 양산 공정 통합

Chris Scanlan · Besi



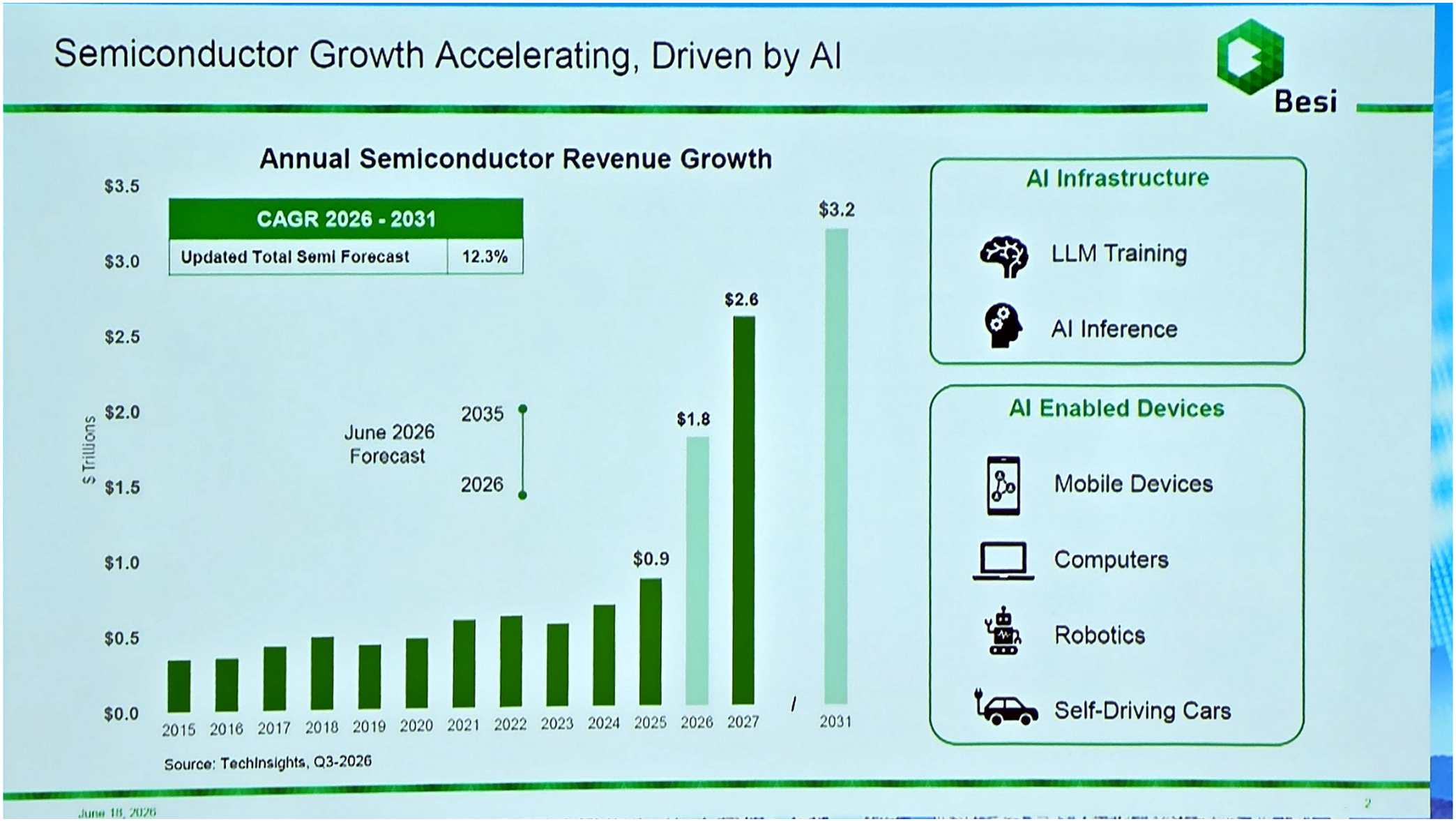
T20 하이브리드 본딩의 양산 공정 통합

Besi · Chris Scanlan



T20 하이브리드 본딩의 양산 공정 통합

Besi · Chris Scanlan



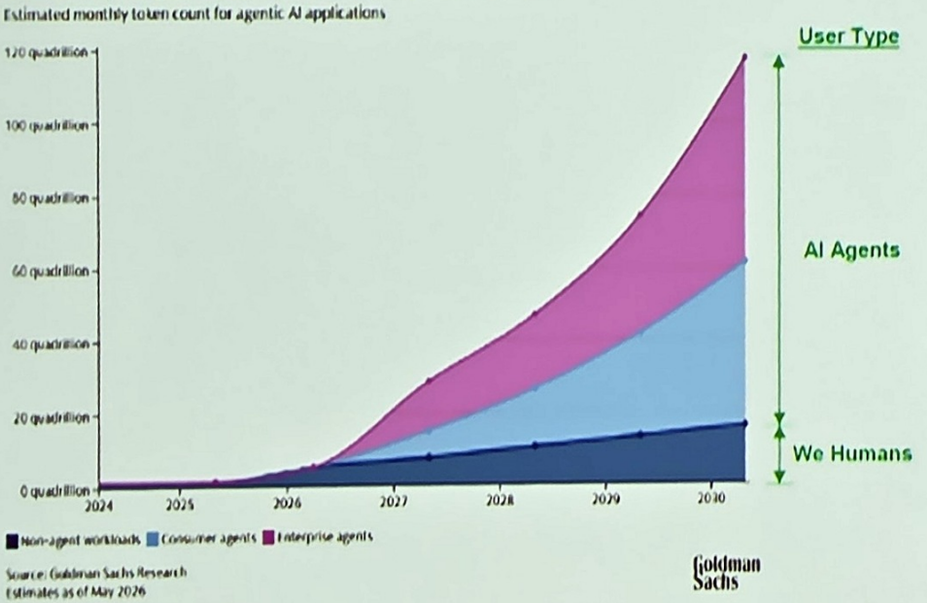
Rapid AI Deployment and Market Adoption Require Large AI Infrastructure Investment by Hyperscalers

Besi

AI model performance and usage growing exponentially

Token use by AI agents is expected to multiply 24 times by 2030

Estimated monthly token count for agentic AI applications



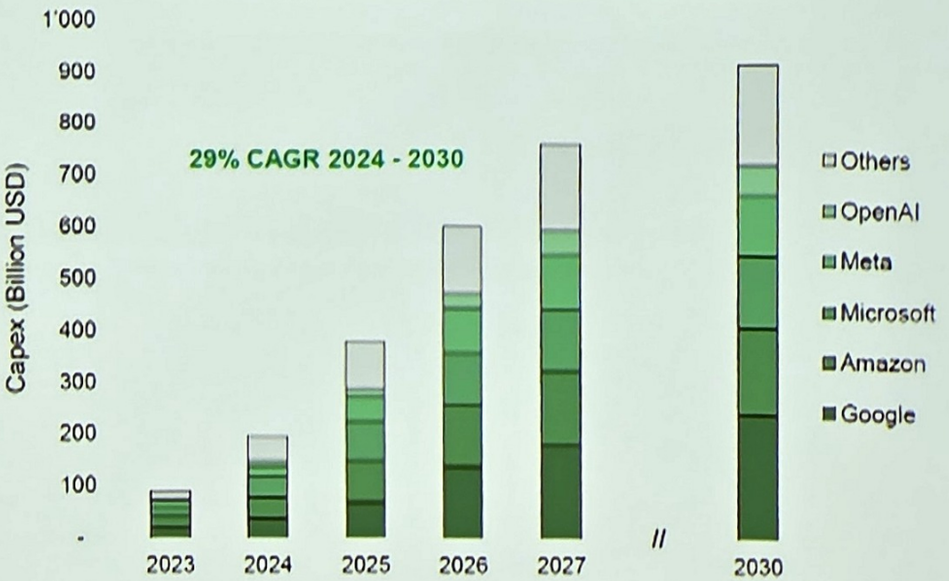
Source: Goldman Sachs Research
Estimates as of May 2026

Goldman Sachs

- Generative AI adopted faster than any previous technology
- Automated AI agents accelerate AI token usage
- Human duration of software development tasks that AI can do successfully is doubling every 7 months (METR.org 2026)

Requiring big investment in computing infrastructure

Annual Hyperscaler Data Center Capex ($B)



Source: TechInsights Q2 2026

- AI infrastructure spending to exceed $1T by 2029 (IDC. April 2026)
- Global AI computing capacity is doubling every 7 months (EPOCH AI, Jan 2026)
- Data centers are projected to represent 3% of global energy consumption by 2030 (Gartner Nov 2025)

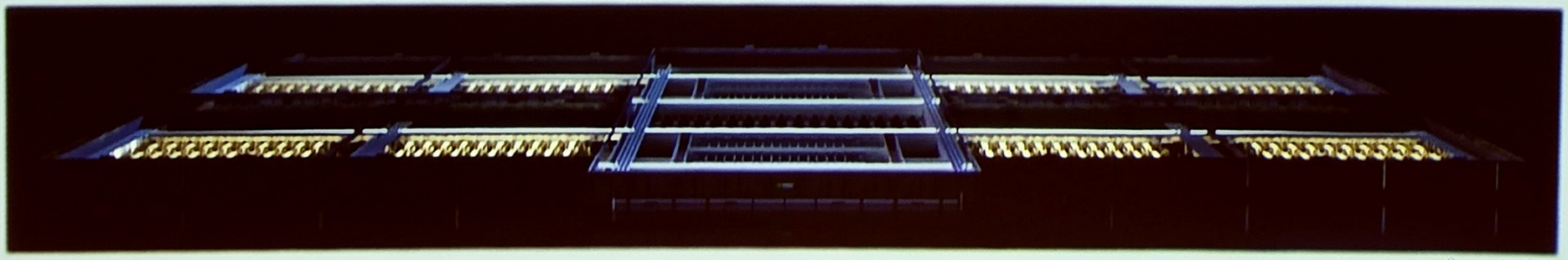
June 18, 2026

3

AI Infrastructure Build-out Drives Demand for High Performance Computing Devices Using Advanced Packaging

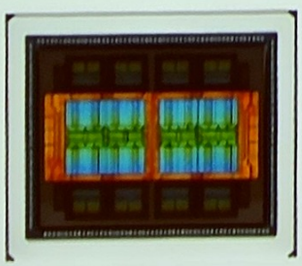
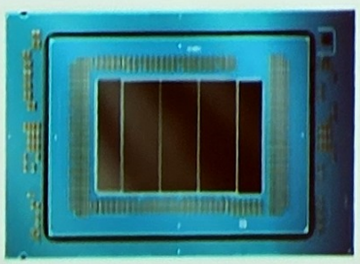
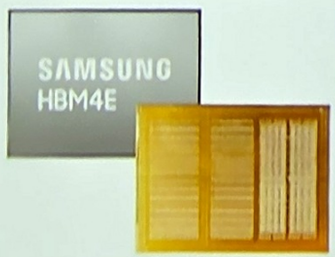
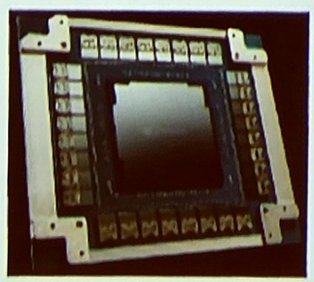
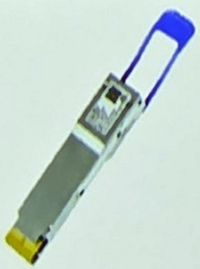
Besi

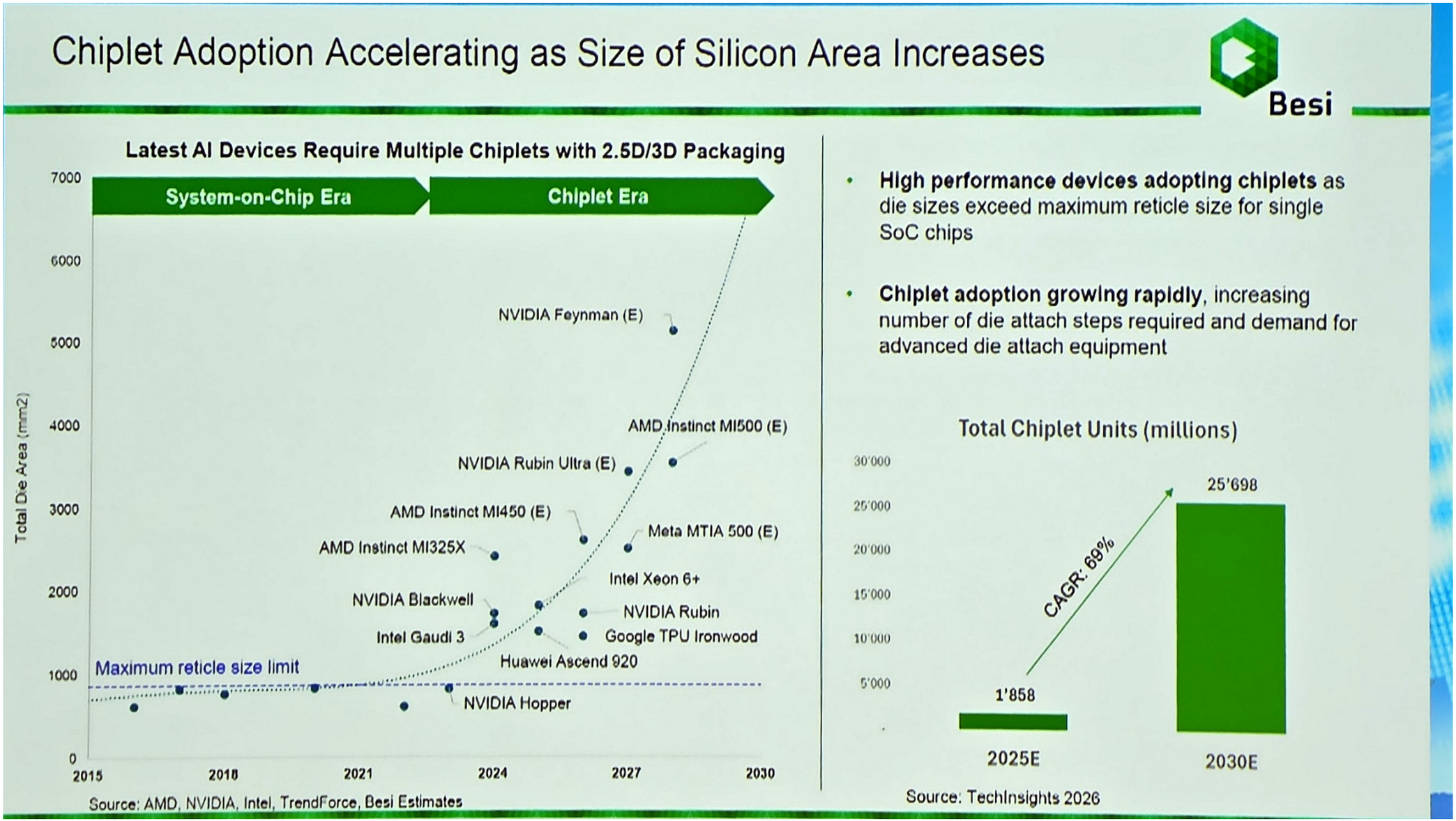
New **AI Factories** link thousands of GPU servers together to make **entire data centers function as one computer**



Source: NVIDIA

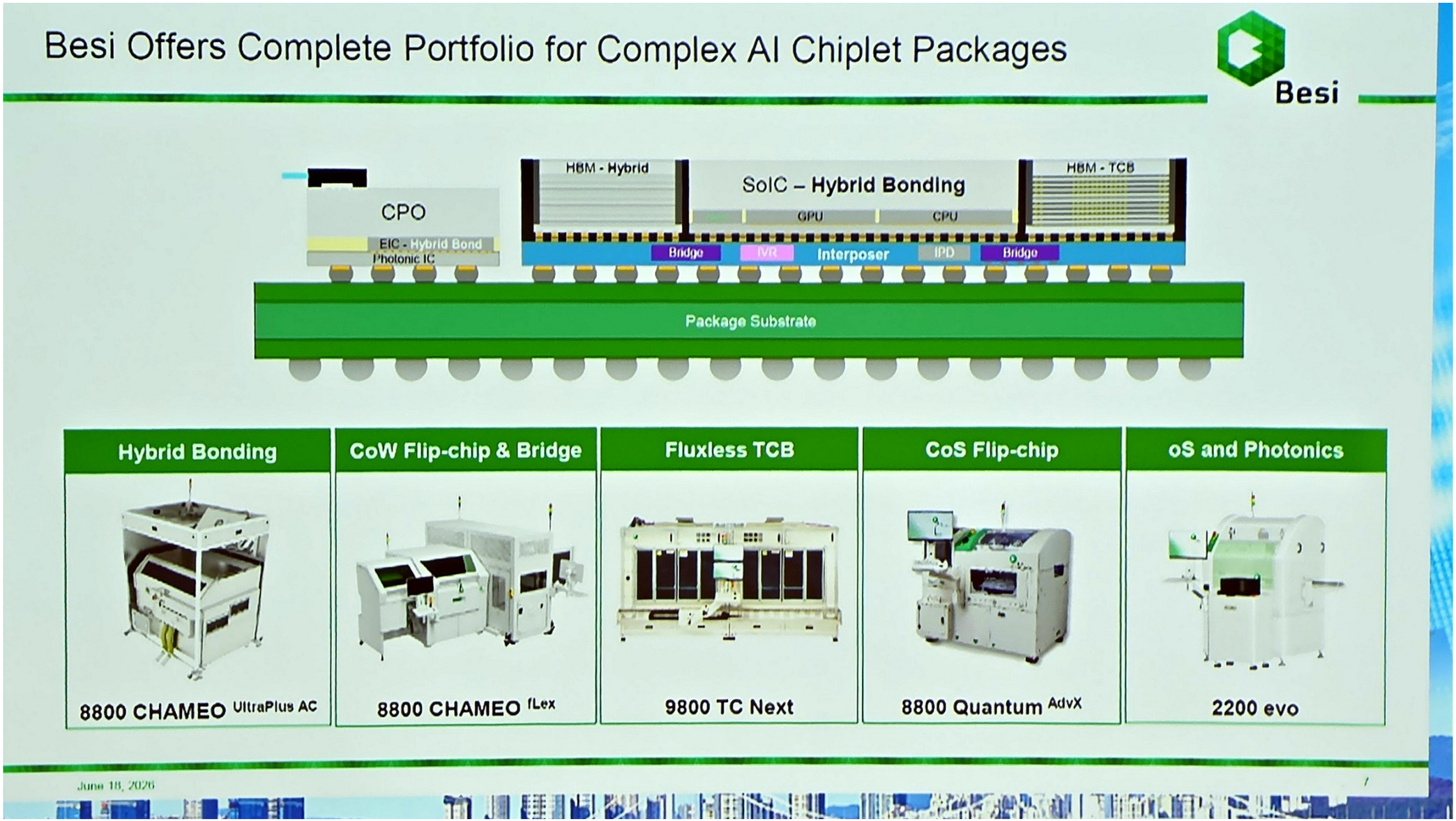
Which drives significant demand for high performance processors, memory and networking devices **using advanced 2.5D & 3D packaging**

GPUs & Accelerators	CPUs	HBM Memory	CPO Network Switches	Photonic Transcievers
 Source: AMD AMD MI355x Accelerator	 Source: Intel Intel Xeon 6+ CPU	 Source: Samsung HBM4E	 Source: NVIDIA NVIDIA Spectrum X Switch	 Source: NVIDIA Ethernet Transceiver


Chiplet Adoption Accelerating as Size of Silicon Area Increases
Besi
Latest AI Devices Require Multiple Chiplets with 2.5D/3D Packaging
System-on-Chip Era
Chiplet Era
Total Die Area (mm2)
7000
6000
5000
4000
3000
2000
1000
0
2015
2018
2021
2024
2027
2030
NVIDIA Feynman (E)
AMD Instinct MI500 (E)
NVIDIA Rubin Ultra (E)
AMD Instinct MI450 (E)
Meta MTIA 500 (E)
AMD Instinct MI325X
Intel Xeon 6+
NVIDIA Blackwell
NVIDIA Rubin
Intel Gaudi 3
Google TPU Ironwood
Huawei Ascend 920
Maximum reticle size limit
NVIDIA Hopper
Source: AMD, NVIDIA, Intel, TrendForce, Besi Estimates
High performance devices adopting chiplets as die sizes exceed maximum reticle size for single SoC chips
Chiplet adoption growing rapidly, increasing number of die attach steps required and demand for advanced die attach equipment
Total Chiplet Units (millions)
30'000
25'000
20'000
15'000
10'000
5'000
25'698
CAGR: 69%
1'858
2025E
2030E
Source: TechInsights 2026

T20 하이브리드 본딩의 양산 공정 통합

Besi · Chris Scanlan

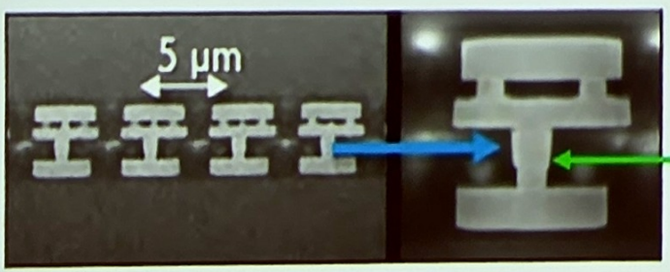


Hybrid Bonding is Interconnect Technology of Choice for High Performance 3D Heterogeneous Integration

Besi

Hybrid Bonding is extension of front-end fab process

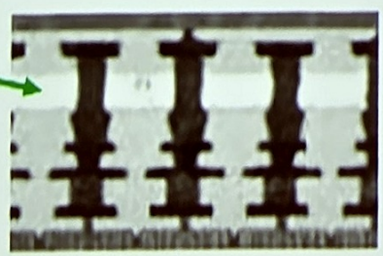
Directly connects chips in 3D with highest interconnect density



Source: AMD

TCB uses µ-bump interconnect to connect chips with solder and epoxy encapsulation materials

Solder µbumps with gap fill material



- **Lower bump density**
- **Higher thermal resistance**
- **Back-end assembly process**

Source: Intel

Hybrid Bonding Benefits for 3D Chip Stacking

- ✓ 15X Higher Interconnect Density[1]
- ✓ 11.9X Higher Speed[1]
- ✓ 191X Higher Bandwidth Density[1]
- ✓ >100X Higher Energy Efficient Performance[2]
- ✓ Up to 30% Lower Thermal Resistance[3]
- ✓ Higher Reliability Performance[4]
- ✓ 10X Lower Cost per Interconnect[4]

Notes: 1) TSMC, 2023 ECTC, 2) TSMC 2025 ECTC, 3) Samsung 2026 ECTC, 4) Besi Company Analysis

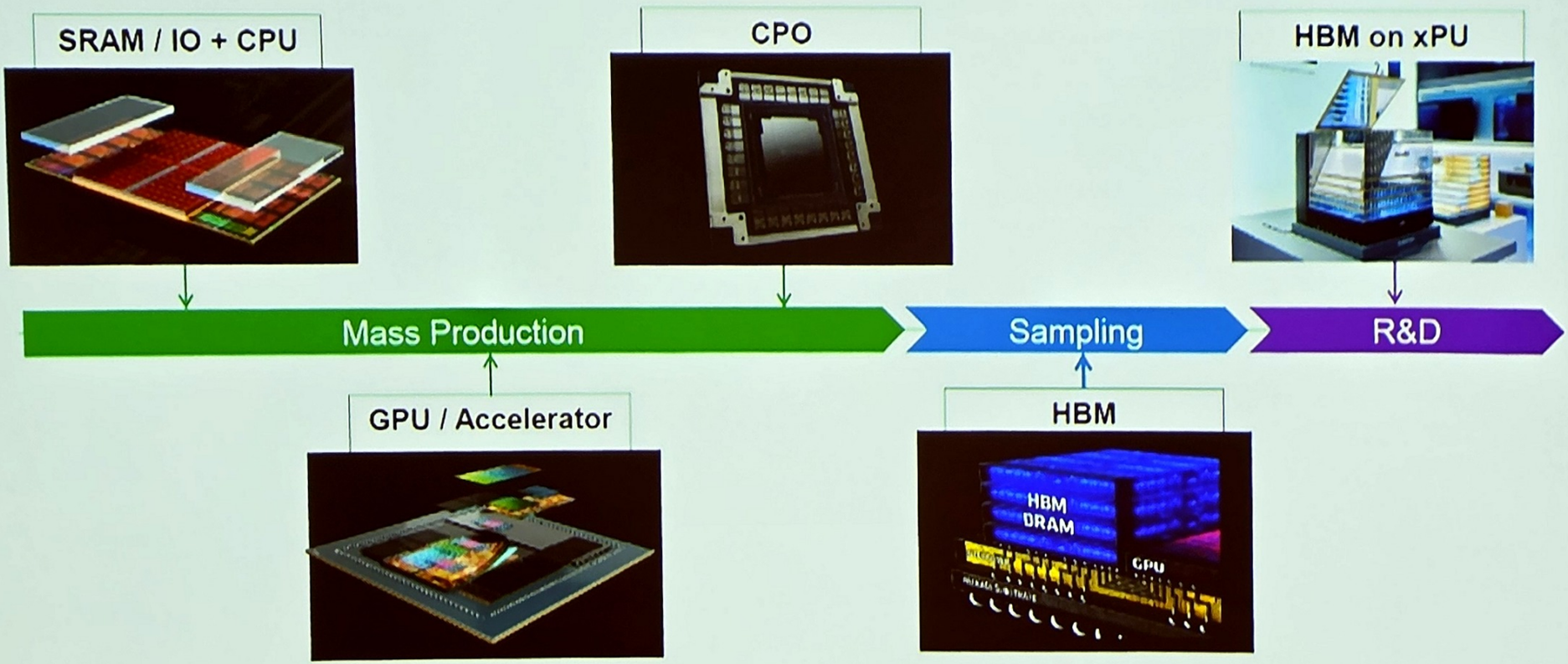
T20 하이브리드 본딩의 양산 공정 통합

Besi · Chris Scanlan

Hybrid Bonding Applications in AI Systems

Besi

Hybrid bonding is used accross compute, memory and networking functions in AI systems



Hybrid Bonding Solves SRAM Bottleneck in Data Center CPUs

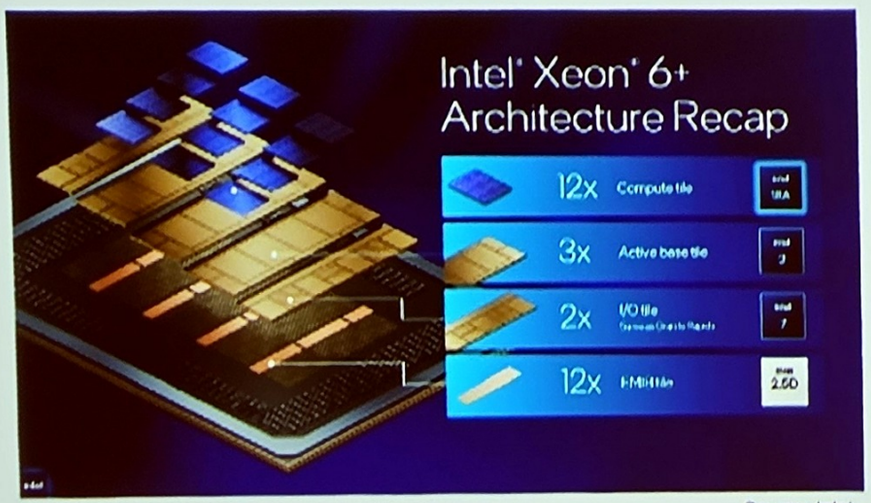
Besi

More SRAM capacity is needed to improve performance of both CPUs and AI Accelerators

Not cost effective to use more advanced nodes for SRAM as performance does not scale below 5nm node

Hybrid bonding solves problem by stacking advanced node core chiplets onto lower cost, trailing node SRAM chiplets

Intel Xeon 6+ Data Center CPU

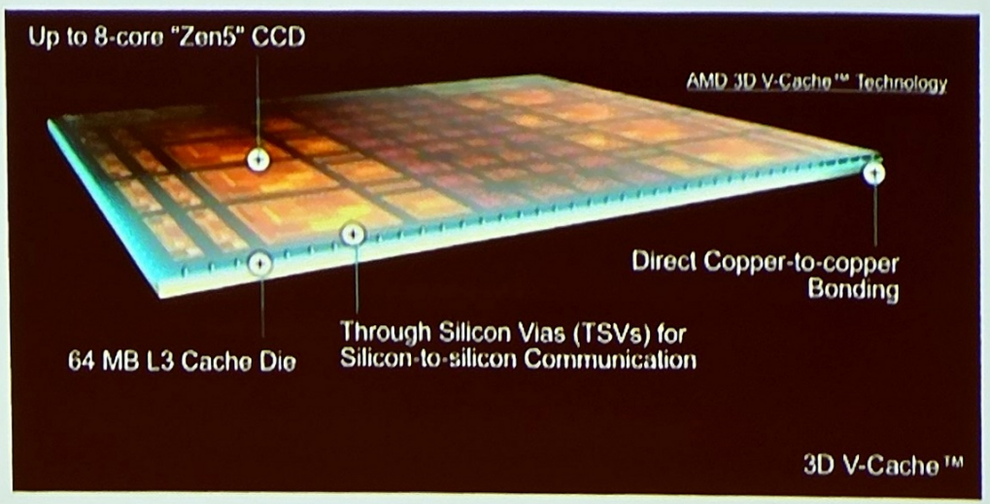


Source: Intel

CPU Core Function Design in Most Advanced 18A Node

- Hybrid bonded onto base tiles with SRAM and other functions
- Limits expensive advanced node usage only to critical core compute

AMD CPUs with 3D V-Cache



In production for desktop and server CPUs

- 3X better interconnect energy efficiency (vs. µbump 3D)
- >15X higher interconnect density (vs. µbump 3D)

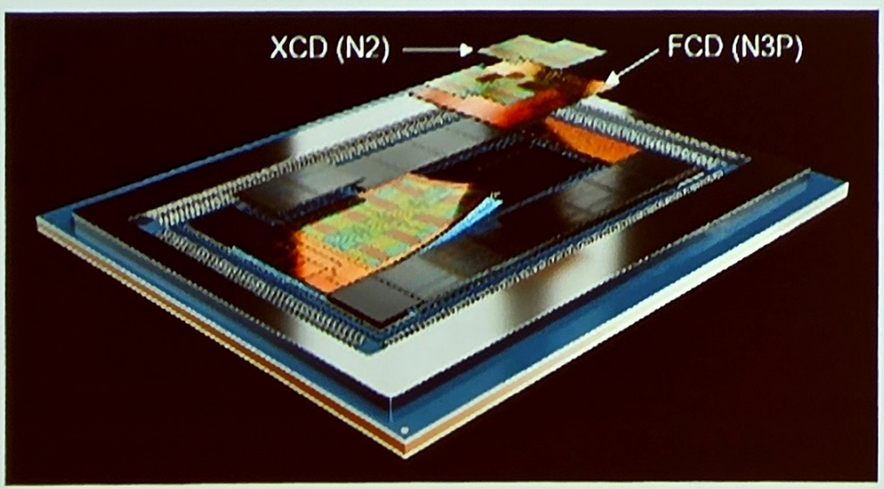
Increased Use of Hybrid Bonding in GPUs and Accelerators Expected

Besi

AMD expanding use of hybrid bonding in AI GPUs

- MI355X with 6 µm pitch hybrid bonding now in HVM
- MI455X with larger die sizes introduced
- Stacking of XCD accelerator chiplets to FCD (I/O Fabric and Cache memory) die using hybrid bonding
- Higher compute density and performance per watt

AMD MI455X GPU



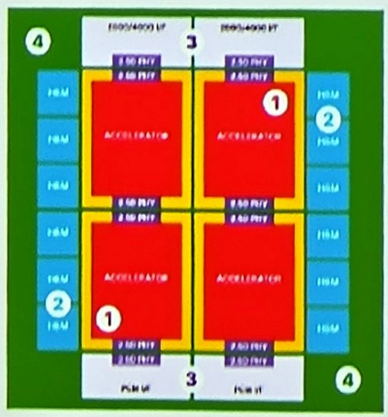
Source: AMD



Source: NVIDIA

NVIDIA's next generation Feynman (2028) has multiple potential hybrid bonding use cases

- Feynman GPU with 3D die stacking
- Introduction of customized HBM



Source: Broadcom

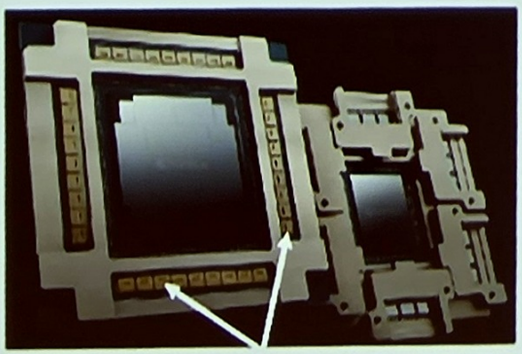
Broadcom's 3.5D XDSiP

- Hybrid bonding of accelerator chiplets to I/O and SRAM die
- Accelerators also benefit from additional SRAM capacity and bandwidth

Hybrid Bonding Becoming Standard for CPO Chiplets in AI Factories

Besi

Co-Packaged-Optics based on TSMC's COUPE with hybrid bonding now in HVM in leading scale-out network switches



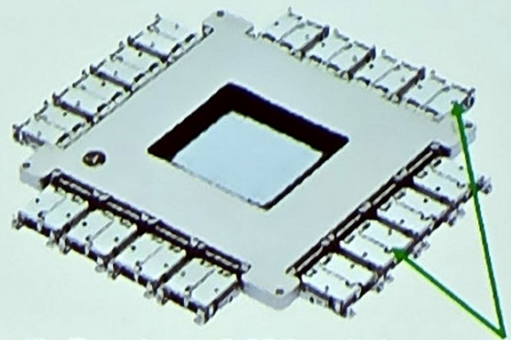
NVIDIA's CPO switches using COUPE

Source: NVIDIA

CPO performance significantly higher compared to traditional pluggable transceivers*:

- ✓ 3.5x power efficiency
- ✓ 10x higher resiliency
- ✓ 1.3x faster installation time

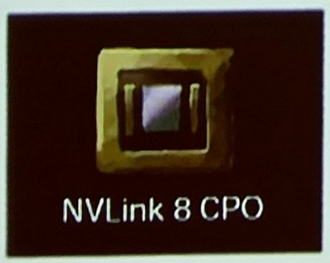
*NVIDIA, 2025 Hotchips



Marvell's Tomahawk 6 CPO switch using COUPE

Source: Marvell

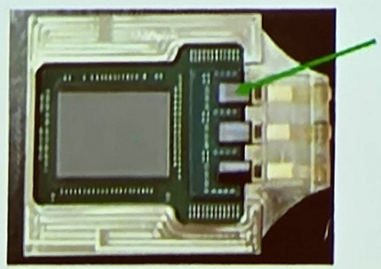
Increased hybrid bonding potential in scale-up networking. CPO will directly connect GPUs to each other within the rack



Source: NVIDIA

NVIDIA's Feynman platform adds CPO to connect GPUs

NVLink 8 CPO is scale-up switch connecting multiple GPUs, replaces copper wiring



Source: AMD

CPO chiplets integrated in GPU package

- ✓ Higher bandwidth
- ✓ Lower power
- ✓ Lower latency

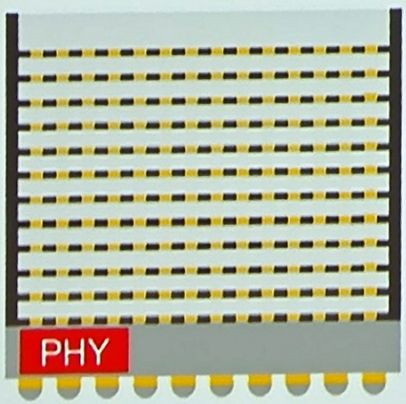
Hybrid Bonding is the Next Scaling Lever for HBM

Besi

HBM is running out of pitch, height and thermal headroom. Direct Cu–Cu links remove the solder-bump penalty

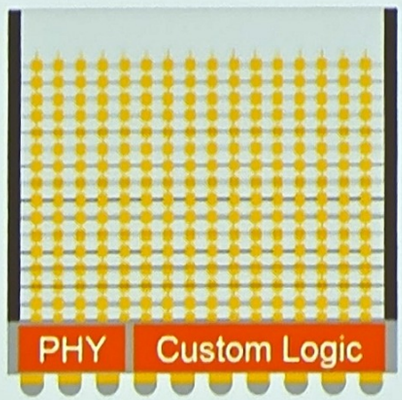
Net benefit: more HBM bandwidth and capacity per package — at lower energy per bit.

Current: HBM with µbump



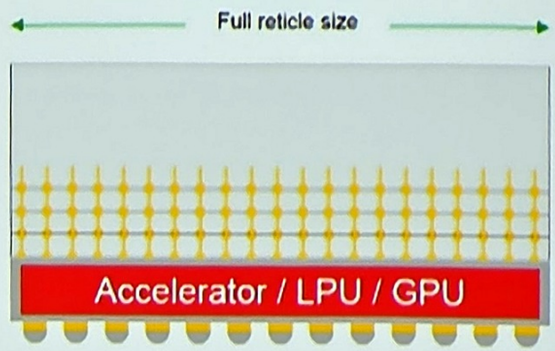
Base die with memory Phy
Limited thermal performance
Limited vertical scaling

Next: HBM with Hybrid Bonding



Base die with custom logic
30% better thermal performance
< 10 µm bond pad pitch

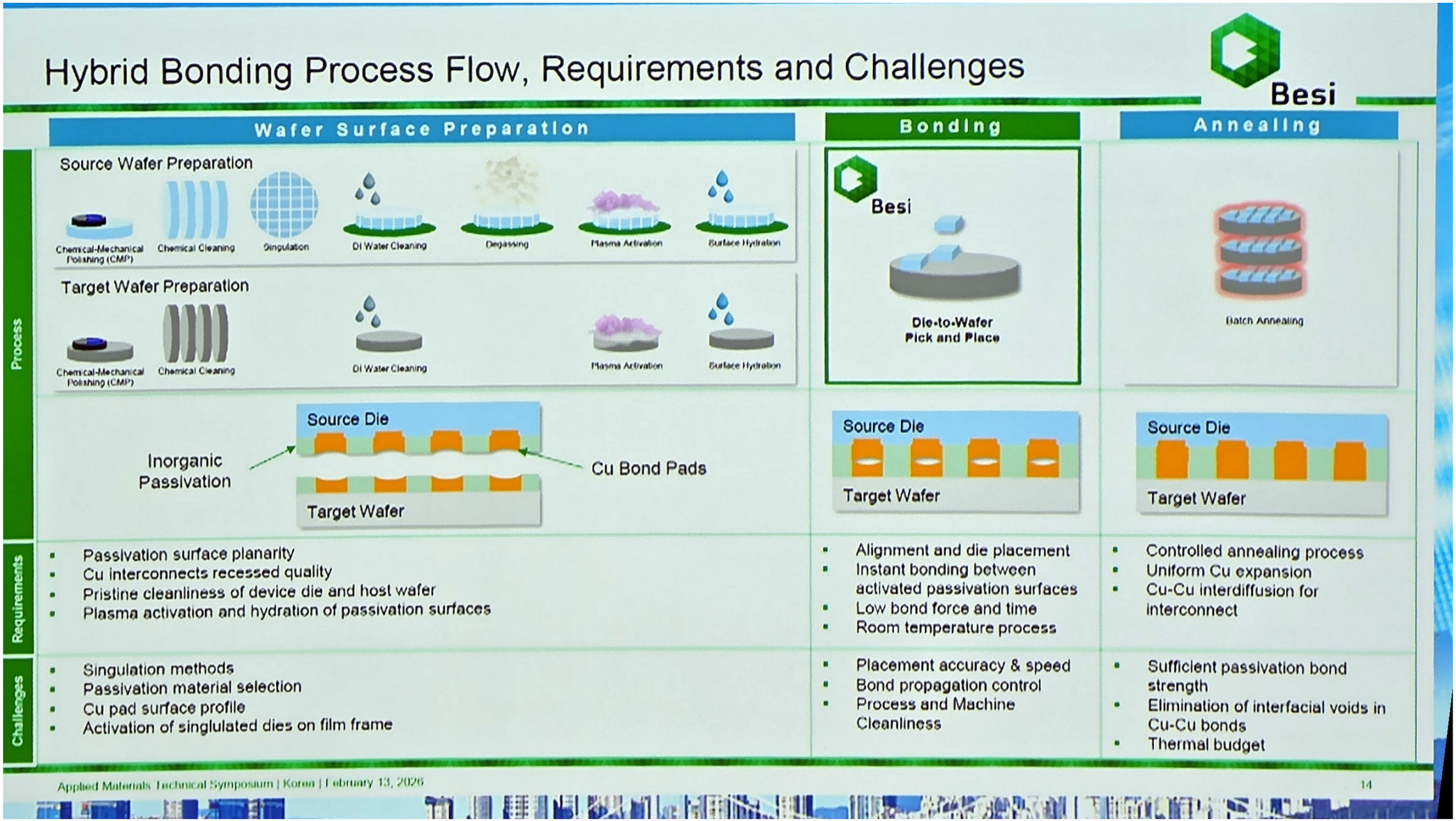
Future: HBM on xPU



HBM stacked directly to GPU
Thermal performance is critical
< 1 µm to 6 µm bond pad pitch

T20 하이브리드 본딩의 양산 공정 통합

Besi · Chris Scanlan



T20 하이브리드 본딩의 양산 공정 통합

Besi · Chris Scanlan

3300 Hybrid *N50* Highest Accuracy and Lowest CoO

Besi

Status: In development – Alpha qualification

0-series Launch: Q3/Q4 2027



3300 HYBRID *N50*

ACCURACY (Alignment)

50 nm (3σ)

Enables < 3 µm pitch hybrid bonding with high yield

THROUGHPUT (Lower CoO)

Up to 3000 CPH

Increases CPH with smaller footprint from AC baseline

PRECISION ARCHITECTURE

Integrated Metrology

Advanced process control and defense → Higher yield at tighter pitches

INTEGRATION BASE

>200 hybrid systems shipped

Accelerates time to POR by leveraging existing automation and data services